

Corporate Credit Risk Modeling

Project Summary

This project implements an IRB-inspired corporate credit-risk modeling framework, for borrower-level PD estimation and calibration, internal rating-band construction, and IFRS 9-style lifetime Expected Credit Loss (ECL) calculation.

It uses a synthetic two-cohort corporate dataset with a time-split validation design: the 2023 cohort is used for model training, while the 2024 cohort is used for out-of-sample testing, calibration, rating-band validation, and ECL estimation.

The modeling pipeline covers: logistic regression with Weight of Evidence (WoE) feature engineering, Prior Correction for probability calibration under severe class imbalance, rating-band design with statistical validation, 12-month and lifetime ECL calculation, multi-period PD projection using three approaches (CAR, matrix powering, and S&P transition-rate anchoring), stochastic LGD and EAD modeling, Gaussian copula dependence modeling for the Monte Carlo simulation of lifetime ECL.

The framework connects three layers of credit-risk analytics: 1) statistical credit risk: logistic-WoE PD estimation and calibration; 2) rating-system design: monotonic rating bands validated by confidence intervals, O/E ratios, and chi-square diagnostics; 3) lifetime credit loss projection: multi-period PD migration, stochastic LGD, EAD projection, and dependence modeling for ECL simulation.

The results show that AUC of base model is 0.777, WoE-logistic is 0.806, getting improved, bootstrap mean is 0.810, with a low standard deviation, out-of-sample is 0.776. Pre-calibration ECE is 0.325, and post-prior-correction is 0.015, showing excellent calibration. Expected lifetime ECL is 154,267. 95% upper tail is 566.910. The distribution of simulated lifetime ECL indicates classic features of credit risk Monte Carlo output, with right-skewed shape, mass near zero, and occasional large-loss scenarios.

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1. Information of synthetic corporate dataset and base model

Chatbot creates a 2-year synthetic corporation dataset for credit risk modeling and analysis. The features and statistics are as follow:

Variable	Description
firm_id	Unique firm identifier
year	Origination year (2023 or 2024), treated as two annual origination cohorts
total_assets	Lognormal firm size proxy
debt_assets	Leverage ratio
roa	Return on assets
current_ratio	Liquidity measure
interest_coverage	Debt servicing capacity
revenue_growth	Growth rate
volatility	Risk proxy
pd	One-year Probability of Default (structurally modeled)
default	Bernoulli-sampled default outcome (0/1)
lgd	Loss Given Default
ead	Exposure at Default

	2023				2024			
	Mean	Std	Min	Max	Mean	Std	Min	Max
total_assets	5,525,415	7,590,950	127,865.30	165,790,900	5,339,781	6,673,456	122,008	78,879,660
debt_assets	0.29	0.16	0.05	0.83	0.29	0.16	0.05	0.90
roa	0.05	0.04	-0.10	0.21	0.05	0.04	-0.10	0.19
current_ratio	1.49	0.50	0.20	3.31	1.50	0.49	0.20	3.22
interest_coverage	3.70	3.31	0.17	35.60	3.75	3.74	0.13	50.00
revenue_growth	0.04	0.10	-0.31	0.41	0.04	0.10	-0.41	0.36
volatility	0.25	0.10	0.05	0.62	0.25	0.10	0.05	0.61
pd	0.03	0.03	0.00	0.16	0.04	0.04	0.00	0.24
lgd	0.45	0.15	0.05	0.95	0.45	0.15	0.05	0.95
ead	1,925,161	3,038,560	21,117.15	89,712,030	1,895,880	2,761,872	17,863	37,211,870

	2023	2024
Non-default - 0	3392	3365
Default - 1	108	135

Total firm number	3500	2500
Average PD	3.09%	3.86%

Such default events yielded from this dataset is suitable for robust modeling, enabling Probability of Default (PD) calibration and rating band validation.

This project uses time-split validation, namely training on 2023 cohort, and testing on 2024 cohort.

The following plots (refer to Appendix) and statistics table show the distributions and significances of each feature for default and non-default respectively on 2023 cohort.

The kernel density estimation plots show that features of total_assets and revenue_growth have no remarkable difference between default and non-default groups. Histogram plots show features of roa, current_ratio, and revenue_growth to be close to normal distributions, and interest_coverage and debt_assets to be lognormal.

Feature	Default_0 Mean	Default_1 Mean	Difference	T-Statistic	P-Value	Effect_Size_Cohens_d
interest_coverage	3.7521	2.1471	1.6050	11.463	0.0000***	0.6312
volatility	0.2495	0.3106	-0.0611	-7.5683	0.0000***	-0.6679
debt_assets	0.2835	0.3438	-0.0603	-4.1806	0.0001***	-0.3972
roa	0.0512	0.0382	0.0130	3.6304	0.0004***	0.3397
current_ratio	1.4951	1.3963	0.0988	1.9967	0.0483*	0.1966
revenue_growth	0.0371	0.0487	-0.0115	-1.1862	0.2380	-0.1155
total_assets	5521622	5644533	-122911	-0.1504	0.8807	-0.0154

Based on t-statistics, the features of interest_coverage, volatility, debt_assets, roa, and current_ratio show statistical significance between default and non-default groups, whereas revenue_growth and total_assets are of no significant differences, which is consistent with the situations indicated in Kernel density estimation plots.

The feature correlation matrix (refer to Appendix) shows there is no high correlations between any two of them.

According the basic analysis of dataset, this project starts to choose logistic regression on raw data. Logistic model is interpretable and regulatory friendly. The following is the concrete settings and approaches.

Regularization	L2, keeping all features
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Hyperparameter tuning approaches	grid search
Cross-validation	stratified K-fold

Given the prominent property of credit dataset, the class imbalance, which suggests minority class carries much more informational value, standard K-fold splitting may result in unstable coefficient estimation in training folds, and highly variable AUC or accuracy. Hence, the regression uses stratified K-fold cross-validation, which can stabilize estimated coefficients and predicted probabilities.

The best results of the base model and hyperparameter tuning show as follow, scoring with AUC:

CV AUC:	0.7772
Test AUC:	0.7719

Cross-validation AUC is 0.777, which serves as the baseline.

2. Feature engineering

2.1 Monotonicity Check

To do feature engineering, this project first checks the monotonic trends of PD and each feature.

The formula: $PD = f(\text{features})$, if function f is non-linear, logistic regression cannot model it. It needs to apply quantile binning and Weight of Evidence (WoE), the engineering techniques, which are powerful for non-monotonic skewed distributions. Concretely, the assumption of logistic regression is the linear relationship between features and the log-odds of the targets.

$$\sigma(z) = \frac{1}{1+e^{-z}}, z = \alpha X + \beta$$

where, σ is logistic regression, and z is the linear form, where X is the vector of features. So, $z = \log \frac{p}{1-p}$, which shows linear-in-log-odds.

If their relationship does not satisfy this assumption, features need to be transformed via quantile binning and WoE, which may handle non-linearity, improve model interpretability and enhance predictive power as well.

The procedure regarding selective feature treatment is as follow:

- 1) Adding quantile bin for each feature
- 2) Computing PD per bin, which is the observed default rate in each bin. The formula is $observed\ PD = \frac{number\ of\ default\ in\ bin\ i}{number\ of\ observations\ in\ bin\ i}$
- 3) Plotting PD vs feature deciles for diagnosis
- 4) If plots are non-monotonic, oscillatory, U-shape, or multiple peaks, these features need to be transformed via quantile binning and WoE

According to plots of PD and each feature by quantile bins (refer to Appendix), the results of feature treatment is listed as follow:

Feature	Use original numerical form or not	WoE needed or not	Explanation
debt_assets	Yes	No	Mostly monotonic, small dip acceptable, upward
interest_coverage	Yes	No	Strongly monotonic, smooth, upward
volatility	Yes	No	Clearly monotonic, downward
debt_assets_roa	REMOVE (interaction term)	—	Unstable, non-monotonic, oscillatory, likely overfitting, may add collinearity
total_assets	No	Yes	oscillatory, noisy
roa	No	Yes	Reversing slope, non-monotonic
current_ratio	No	Yes	Highly irregular
revenue_growth	No	Yes	Non-monotonic

2.2 Weight of Evidence (WoE) Transformation

Next part is to apply WoE to features, total_assets, roa, current_ratio, and revenue_growth.

WoE is to transform categorical, binned continuous variables into a single continuous variable that measures the “strength” of each category in predicting the binary outcome.

The formula is $WoE = \ln \left(\frac{percentage\ of\ default}{percentage\ of\ non-default} \right)$

For each bin:

$$\text{percentage of default} = \frac{\text{number of default in bin } i}{\text{total number of default across all bins}}$$

$$\text{percentage of non - default} = \frac{\text{number of non - default in bin } i}{\text{total number of non - default across all bins}}$$

WoE > 0	the category has higher proportion of default
WoE < 0	the category has higher proportion of non-default
WoE ≈ 0	not very informative

Also, WoE is often paired with Information Value (IV).

The formula is

$$IV = \sum_i \left(\frac{\text{number of default in bin } i - \text{number of non - default in bin } i}{\text{total non - default across all bins}} * WoE_i \right)$$

across all bins

According to the requirements of Basel rules, FICO scorecards, and IFRS9, it needs to check non-monotonic pattern of PD and WoE, before WoE encodes.

The specific procedure is as follow:

- 1) Computing WoE per bin, and IV
- 2) Checking whether WoE is non-monotonic
- 3) If non-monotonicity, calculating Chi-square for two adjacent bins, and merging them. Merging criteria: violation of monotonicity or smallest Chi-square value
- 4) Repeating the steps above, until monotonicity is achieved (stop condition). Safety condition: setting maximum iteration and minimum number of bins

The results of WoE and IV is listed below:

Feature	WoE: monotonic pattern is achieved or not	Initial IV	Final IV	
total_assets	Yes	0.8277	0.0041	IV<0.02, Drop
roa	Yes	1.7512	0.8326	WoE value

current_ratio	Yes	1.0294	0.1150	WoE value
revenue_growth	Not yet	1.5892	0.0094	IV<0.02, Drop

As for features, roa and current_ratio, it needs to check the linearity of WoE value and log-odd. The plots (refer to Appendix) show monotonicity for both of them.

Therefore, the final verdict on feature treatment is as follow:

Feature	Treatment for logistic regression
debt_assets	Original data
interest_coverage	Original data
volatility	Original data
roa	WoE value
current_ratio	WoE value
debt_assets_roa	Drop
total_assets	Drop
revenue_growth	Drop

Based on the feature engineering, finally there are five features in logistic regression, and two of them use WoE values. The elimination of current_ratio and revenue_growth is consistent with the result of t-test, that is these two features show no statistical significance.

3. Regression on WoE transformation

3.1 Regression Results and Validation

The regression result show that the best cross-validation ROC-AUC score is 0.8062. It is acceptable and a good discrimination. Also, it is higher than 0.7777 of base model.

Classification report:

	Precision	Recall	F1-score	Support
0	0.99	0.69	0.81	3392
1	0.07	0.74	0.13	108
accuracy			0.69	3500
macro avg	0.53	0.71	0.47	3500
weight avg	0.96	0.69	0.79	3500

Confusion Matrix

2333	1059
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Recall of 74% is very strong. Low precision is normal for credit risk, such as PD is around 3%. F1-score is meaningless for imbalanced credit datasets. F1-score equally weights precision and recall, but in credit risk, missing a default is very costly, whereas false alarms are less costly.

3.2 Bootstrap Analysis for Diagnostics

To align with Basel and IFRS9, it needs to do diagnostics, checking the model validation and reliability before the calibration. This project implements bootstrap for checking coefficient sign stability, via fixing the best model, and then refitting with fixed hyperparameters for each bootstrap.

Based on 1000 bootstrap samples, the result is listed below, as well as coefficient intervals plots, coefficient distribution plots and AUC distribution (refer to Appendix).

Coefficient Confidence Intervals (95%)				
Feature	Bootstrap_Mean	Bootstrap_Std	CI_Lower_95%	CI_Upper_95%
roa_woe	1.131259	0.312440	0.545217	1.771222
current_ratio_woe	1.106805	0.346509	0.465041	1.813268
debt_assets	0.518005	0.106937	0.304184	0.726090
interest_coverage	-1.456988	0.220958	-1.920334	-1.064021
volatility	0.795075	0.105753	0.602330	1.012870
Intercept	-0.809821	0.130770	-1.084734	-0.582442

Model Stability (ROC-AUC)	
Mean AUC	0.8100
Std AUC	0.0165
95% CI_lower	0.8422
95% CI_higher	0.7763
Range width	0.0659

All confidence intervals exclude 0, keeping the same sign across all bootstrap samples. It is a strong statistical and practical significance. Also, all signs are consistent with economic meanings. Standard deviation values are small.

Bootstrap coefficient distribution plots show variance of each coefficient, and their distributions are roughly symmetric, with no remarkable tail behavior.

Mean AUC 0.81 is a strong performance for a PD model in credit risk. Std AUC 0.0165 < 0.02 is a very low volatility, showing stable and reliable, and indicating good generalizability and no material overfitting. Range width 0.066 is narrow, indicating the high stability.

According to analysis above, it is a robust logistic-WoE credit modeling.

4. Calibration

Evaluation on the out-of-sample dataset (2024) results in statistic values and various plots (refer to Appendix).

Brier Score	0.1935
Expected Calibration Error (ECE)	0.3251
ROC-AUC (discrimination)	0.7755
2024 Actual default rate	3.8571%
2024 Mean predicted probability	36.3629%

AUC is 0.7755 > 0.7, which indicates model has good discrimination, but ECE is 0.3251 > 0.1, reflecting poor calibration, namely predicted probabilities do not match actual rates. 2024 mean predicted probability 36.3629% shows the model is over-predicting defaults significantly. Also, the calibration curve indicates model probabilities systematically too high.

Given the situations of strong imbalance class (default rate is 3%-4%), WoE transformation, small dataset (3500 observations), and distribution shift (PD model on one year training and another one year applying), it is the expected pattern.

Due to the extreme over prediction, it needs to perform downsampling on the majority class (non-defaults) during training to handle class imbalance. The project applies the "Prior Correction" formula.

The formula is:

$$\beta = \frac{\pi_{true}}{1 - \pi_{true}} * \frac{1 - \pi_{train}}{\pi_{train}}$$

$$P_{corrected} = \frac{P_{original} * \beta}{P_{original} * \beta + (1 - P_{original})}$$

β : correction factor

$P_{original}$: model's predicted probability (trained with class weights)

π_{true} : true default probability in test set, 0.03857

setting class_weight = 'balanced' in training

For balanced weight, $\frac{1-\pi_{train}}{\pi_{train}} = 1$

The result is showed as follow, and the plots as well (refer to Appendix).

True pred:	3.8571%
Original pred:	36.3629%
Corrected pred:	4.2221%

After Prior Correction, the project applies Platt scaling on top of the prior-corrected probabilities. Comparing Platt scaling, Beta calibration, and Isotonic regression, Platt scaling has lower risk of overfitting, whereas Isotonic regression has higher risk of overfitting.

	Fit 2024 dataset before calibration	Prior correction	Platt scaling
ROC-AUC Score	0.7755	0.7755	0.7208
Brier Score	0.1935	0.0367	0.0363
Expected Calibration Error (ECE)	0.3251	0.0150	0.0218
Predicted probability of default (2024)	36.3629%	4.2221%	
True probability of default (2024)	3.8571%		

The result above shows that ROC-AUC score and ECE become worse, after applying Platt scaling on top of the Prior-Corrected result. Therefore, this project sticks with the Prior Correction result, and it is the most robust, principled and stable solution for this dataset. The specific reason is that remaining calibration error (ECE: 0.015) is already too small to be learned by any model, and any attempt to fit a complex curve to such a small, residual error will capture noise instead of signal.

5. Rating Band Design

5.1 Criteria and Trials of Rating Band

The goals and criteria of rating band design:

- 1) Risk discrimination: each band has a distinct and statistically stable default rate

- 2) Monotonicity across bands
- 3) Reasonable granularity

Steps for rating band

- 1) Using the corrected probability of default (predicted) with quantile to define bands, and setting the initial number of bins
- 2) Calculating true probability of defaults, namely actual realized default rate
- 3) Checking monotonicity based on actual realized default rates and merging the violations according to Chi-square test (selecting minimum Chi-square value iteratively)
- 4) Stop condition: monotonic pattern is achieved; safety conditions: minimum number of bins, and minimum number of defaults per bin

Based on several trials, the project implements the two-phase strategy. First priority is the constraint of minimum number of defaults (reliability), and second one is monotonicity (interpretability), and relaxation of the condition, the minimum number of bins.

As for the consideration of the priority, the safety and stability condition, minimum number defaults per bin, must always take precedence over the desired number of bins. A bin with too few defaults provides an unreliable estimate of the PD-in-band, which invalidates the subsequent ECL.

The core function of a PD model is discrimination and ranking. Monotonicity confirms that final defined risk grades follow the original ranking from the model.

In addition, second trial shows that dataset does not contain enough statistical power to reliably distinguish between 7 distinct risk grades. 7 bins lead to noise that violates the risk hierarchy.

First trial	Second trial	Third trial
Total population: 3500	Total population: 3500	Total population: 3500
Total defaults: 135	Total defaults: 135	Total defaults: 135
Conditions	Conditions	Conditions
Initial bins:15	Initial bins:15	Initial bins:15
Min defaults per bin: 5% of total number of defaults	Min defaults per bin: 5% of total number of defaults	Min defaults per bin: 5% of total number of defaults
Min number of bins: 7	Min number of bins: 7	Min number of bins: 7
Results	Results (first phase)	Results (first phase)

Monotonicity achieved	Minimum number of defaults per bin satisfied	Minimum number of defaults per bin satisfied
Final number of bins: 10	Bins remaining (first phase): 9	Bins remaining (first phase): 9
Minimum number of defaults per bin NOT satisfied	Bins remaining (second phase): 7	Bins remaining (second phase): 6
	Results (second phase)	Results (second phase)
	Final number of bins: 7	Final number of bins: 6
	Minimum number of defaults constrain: Pass	Minimum number of defaults constrain: Pass
	Monotonicity: FAIL	Monotonicity: Pass

The rating bands is showed as follow, also the plots are predicted PD by rating grade, bin composition, life curve, and calibration plot (refer to Appendix).

Grade	PD_Range	Total	Pct_of_T otal (%)	Number_of _Defaults	Calibrated_PD _Mean (%)	Actual_Default _Rate (%)
Band 1	(-0.00100, 0.01690]	1633	46.66	10	0.614	0.612
Band 2	(0.01690, 0.03660]	700	20.00	26	2.588	3.714
Band 3	(0.03660, 0.04710]	233	6.66	12	4.179	5.150
Band 4	(0.04710, 0.09010]	467	13.34	33	6.548	7.066
Band 5	(0.09010, 0.14100]	233	6.66	21	11.020	9.013
Band 6	(0.14100, 0.65600]	234	6.69	33	22.924	14.103

5.2 PD Validation by Rating Band

To check whether expected defaults match realized defaults, including per-band comparison and aggregate portfolio, this project applies three approaches to do validation, Chi-square test across bands (overall discriminatory structure), binomial test per band (band-level calibration), and observed/expected (O/E) ratio per band. Binomial test has two tests: one is exact binomial test (Clopper-Pearson), which is the formal hypothesis testing for statistical significance, and another is Wilson score interval, which is widely used in credit risk analysis for estimation and diagnostics.

The results are listed below:

Wilson score interval							
Grade	PD_Range	CI_Lower	CI_Upper	CI_Width	Predicted_ in_CI	P_value	Significant _5%

Band 1	(-0.00100, 0.01690]	0.003330	0.011236	0.007906	Yes	1.000000	No
Band 2	(0.01690, 0.03660]	0.025471	0.053867	0.028395	Yes	0.072388	No
Band 3	(0.03660, 0.04710]	0.029704	0.087850	0.058146	Yes	0.412801	No
Band 4	(0.04710, 0.09010]	0.050756	0.097577	0.046821	Yes	0.639486	No
Band 5	(0.09010, 0.14100]	0.059705	0.133848	0.074143	Yes	0.401722	No
Band 6	(0.14100, 0.65600]	0.102212	0.191435	0.089222	No	0.001030	Yes

Clopper-Pearson

Grade	PD_Range	CI_Lower	CI_Upper	CI_Width	Predicted_ in_CI	P_value	Significant 5%
Band 1	(-0.00100, 0.01690]	0.002940	0.011233	0.008292	Yes	1.000000	No
Band 2	(0.01690, 0.03660]	0.024404	0.053951	0.029547	Yes	0.072388	No
Band 3	(0.03660, 0.04710]	0.026892	0.088235	0.061343	Yes	0.412801	No
Band 4	(0.04710, 0.09010]	0.049138	0.097810	0.048672	Yes	0.639486	No
Band 5	(0.09010, 0.14100]	0.056659	0.134470	0.077812	Yes	0.401722	No
Band 6	(0.14100, 0.65600]	0.099105	0.192315	0.093210	No	0.001030	Yes

Chi2_statistic	15.539370
P_value	0.008290
Degrees_of_freedom	5

Grade	OE_ratio	OE_light	P_value_wilson	Traffic Light	Final_Traffic_Light
Band 1	0.997345	Green	1.000000	Green	Green
Band 2	1.435195	Amber	0.072388	Green	Amber
Band 3	1.232404	Amber	0.412801	Green	Amber
Band 4	1.079166	Green	0.639486	Green	Green
Band 5	0.817865	Green	0.401722	Green	Green
Band 6	0.615188	Amber	0.001030	Red	Red

The highest-risk rating band exhibits statistically significant PD overestimation. Both exact binomial testing and confidence-interval diagnostics indicate that the calibrated PD materially exceeds observed default experience. Also, Clopper-Pearson and Wilson score have the agreement on band 6, so the conclusion is robust.

Though O/E ratio show the test pass for all bands, combined results retain the overestimation of band 6 with statistical significance.

Chi-square test reject the null hypothesis of perfect calibration. This outcome is consistent with band-level exact binomial testing results. Band 6 failure alone is sufficient to make any aggregated chi-square test reject. However, the miscalibration is localized and does not indicate a structural failure of the rating system.

According to the analysis above, the model is conservative, but it is inefficient, and it may rightly trigger remediations, suggesting potential for recalibration to improve risk sensitivity. Another way is that give the deviation is conservation, this overestimation acts as a conservative buffer, and no immediate model recalibration is required for safety, and it needs to do post-validation ECL overlay analysis and future iterations aiming to reduce this variance.

6. Expected Credit Loss (ECL) calculation

6.1 ECL Calculation Framework

There are generally two frameworks of ECL calculations: IFRS 9 and CECL. IFRS 9 style is an international and 3-stage dynamic model. CECL is US GAAP, static lifetime measurement. This project applies IFRS 9 framework.

12-month ECL: $ECL = \sum_i PD_i * LGD_i * EAD_i$

Lifetime ECL: $lifetime\ ECL = \sum_t marginal\ PD_t * LGD_t * EAD_t * DF_t$

marginal PD_t: PD in period t, given survival to that period, predicted, and calibrated

LGD_t: loss given default, loss fraction, future LGD estimate, may remain constant, or vary by scenarios

EAD_t: exposure at default, loss scale, future EAD projections, typically decreases as loans amortize

DF_t: discounted factor for period t

6.2 12-month ECL

Based on 12-month ECL formula, the 12-month ECL and ECL per exposure is listed below.

	PD_12m_2024	LGD_2024	EAD_2024	ECL_12m
mean	0.04	0.45	1,895,880.00	35,713.26

std	0.06	0.15	2,761,872.00	146,791.60
min	0.00	0.05	17,862.81	0.00
25%	0.01	0.35	473,211.30	1,728.97
50%	0.02	0.45	1,033,688.00	7,880.65
75%	0.05	0.55	2,241,704.00	28,284.49
max	0.66	0.95	37,211,870.00	6,435,221.00

Rating_band	ECL per EAD
1	0.0028
2	0.0117
3	0.0181
4	0.0289
5	0.0477
6	0.1070

The EAD-weighted 12-month ECL increases monotonically from 0.28% in the lowest-risk rating band to 10.70% in the highest-risk-band, confirming strong risk differentiation and ordinal consistency of the rating scale.

6.3 Lifetime ECL

6.3.1 Multi-period PD projection

This project implements three methods to calculate multi-period PD term structure: compounding annual rate (CAR), matrix powering, and S&P rating matrix anchor.

1) Compounding Annual Rate (CAR)

The assumption: default rate p is i.i.d. $\sim$ Bernoulli trials

It is a discrete-time approximation.

Each year: probability of default: $p = PD_{12m}$

Survival per year: $1 - p$

After T years: $P(\text{survival to } T) = (1 - p)^T$

Accumulative: $P(\text{default by } T) = 1 - (1 - p)^T$

The continuous-time limit of CAR is constant hazard rate (CHR), which is the survival-analysis model. Its assumption: default follows a Poisson process, namely default time follows an exponential distribution, with constant intensity λ .

According to Poisson limit theorem, the differences of CAR and CHR is immaterial, if $PD < 20\%$, and horizon < 15 years. There is divergence if $PD > 20\%$ with longer tenors, e.g. horizon of 15-20 years.

For the calculation of ECL, it is suitable to use CAR.

2) Matrix Powering

This project maps rating bands to approximate 1Y PD levels from S&P long-term global corporate default studies. This mapping is used to anchor predicted and corrected PD of 2024 to S&P transition matrix.

Rating_band	PD_Range	S&P Rating
Band 1	(-0.00100, 0.01690]	BB
Band 2	(0.01690, 0.03660]	B
Band 3	(0.03660, 0.04710]	B
Band 4	(0.04710, 0.09010]	B
Band 5	(0.09010, 0.14100]	CCC/C
Band 6	(0.14100, 0.65600]	CCC/C

And then, extracting and recalculating BB, B, CCC/C from S&P one-year transition rate¹ (refer to Appendix), getting 1Y transition matrix

	BB	B	CCC/C	D
BB	0.9152	0.0725	0.0058	0.0065
B	0.0512	0.8605	0.0548	0.0335
CCC/C	0.0047	0.1555	0.5317	0.3081
	0.0000	0.0000	0.0000	1.0000

Based on 1Y transition matrix, calculating nY transition matrix by matrix powering P^n .

Then, extracting last column D and excluding the bottom row. Each column is the cumPD_nY, where $n = 2, 3, \dots$

Next, it needs to anchor S&P transition rates ($S\&P_cumPD_t$) to predicted and corrected PD of 2024 ($PD_internal$), getting $adjusted_cumPD_t$.

Survival power formula: $adjusted_cumPD_t = 1 - (1 - S\&P_cumPD_t)^{scalar}$

$$scalar = \frac{\ln(1 - PD_internal)}{\ln(1 - S\&P_cumPD_1Y)}$$

¹ <https://maalot.co.il/Publications/FTS20250331162126.pdf>

where, $1 - PD_{internal}$ is survival probability of $PD_{internal}$, and $1 - S\&P_cumPD_{1Y}$ is survival probability of $S\&P_cumPD_{1Y}$

$$\text{therefore, } marginal_PD_t = \frac{adjusted_cumPD_t - adjusted_cumPD_{t-1}}{1 - adjusted_cumPD_{t-1}}$$

where, $1 - adjusted_cumPD_{t-1}$ is survival probability_{t-1}

for the first year (1Y), $marginal_PD = adjusted_cumPD_{1Y}$

3) S&P Multi-horizon Transition Rates

Based on average multiyear global corporate transition matrices (refer to Appendix), multi-horizons, 1, 2, 3, 5 years, doing interpolation to derive the 4-year.

Horizon	BB	B	CCC/C
1	0.0056	0.0293	0.2612
2	0.0177	0.0698	0.3579
3	0.0318	0.1060	0.4107
4	interpolation		
5	0.0600	0.1637	0.4818

For interpolation, it needs to do in log space with survival probability, since log of probability of default (PD) does not have linear properties in time.

Log-linear survival interpolation:

$$\ln(survival_prob_4) = \ln(survival_prob_3) + \frac{1}{2}(\ln(survival_prob_5) - \ln(survival_prob_3))$$

where, $survival_prob = 1 - PD$

therefore, it gets the complete cumulative PD matrix (1-5 years).

Then, it needs to anchor S&P cumulative PD ($cumPD_t$) to predicted and corrected PD of 2024 ($PD_{internal}$), getting $adjusted_cumPD_t$.

$$scalar = \frac{\ln(1 - PD_{internal})}{\ln(1 - S\&P_cumPD_{1Y})},$$

$$adjusted_cumPD_t = 1 - (1 - cumPD_t)^{scalar},$$

The calculation of $marginal_PD_t$ is the same as the formula in second approach, matrix powering.

According to three methods above, the marginal PDs for 5 horizons are listed below

CAR	Marginal PD			
Horizon	Mean	Std	Min	Max
1	0.042221	0.062861	0.000000	0.655517
2	0.036488	0.045667	0.000000	0.250000
3	0.032150	0.035564	0.000000	0.148147
4	0.028734	0.028908	0.000000	0.105469
5	0.025963	0.024204	0.000000	0.081920

Matrix powering	Marginal PD			
Horizon	Mean	Std	Min	Max
1	0.042221	0.062861	0.000000	0.655517
2	0.045897	0.051960	0.000000	0.555425
3	0.046035	0.043653	0.000000	0.448116
4	0.045064	0.038223	0.000000	0.352453
5	0.043986	0.035021	0.000000	0.279181

S&P multi-horizon transition rates	Marginal PD			
Horizon	Mean	Std	Min	Max
1	0.042221	0.062861	0.000000	0.655517
2	0.041084	0.037355	0.000000	0.389726
3	0.036428	0.029020	0.000000	0.260719
4	0.031453	0.023560	0.000000	0.202532
5	0.031453	0.023560	0.000000	0.202532

The marginal PDs of CAR show no credit deteriorations, no rating migration, and no downgrade-driven convexity in default risk. The means for 1-5 horizons show strictly decline, smooth and low tail risk. Additionally, their marginal PDs are all less than those from matrix powering, which is expected.

The marginal PDs of matrix powering reflect rating migrations, credit deteriorations, and convex default accumulations. It embedded convexity via downgrades. The Markov

chain allows accelerating default intensity, and captures the dynamic nature of credit risk better than using static, published cumulative rates. Mean PD patterns show initial increasing, and peaks around 2-3 horizons, and slow decreasing. This characteristic “hump” shaped marginal PD profile generated from the time-homogeneous Markov chain on credit state is consistent with empirical default studies and the theoretical predictions of intensity models.

The marginal PDs of S&P multi-horizon transition rate anchor have no migration dynamics, and not a Markov process. Published S&P cumulative PDs may be long-run averages and smoothed across cycles. 4-5 horizons are the same, suggesting hazard flatten effects.

This project chooses the results from matrix powering as the inputs in the ECL calculation, since they indicate migration-driven path, state transitions, endogenous deteriorations and hazard hump.

6.3.2 Loss Given Default (LGD) Estimator

1) Simulation of LGD Behavior

According to the statistical property of LGD data in 2024, this project implements the semi-parametric method, namely parametric core plus directly empirical CDF tail splicing to simulate the LGD.

Mean	0.450385	
Std	0.148080	
Min	0.050000	
Max	0.950000	
Skew	0.024760	near zero: symmetric
Kurtosis	-0.065689	slightly platykurtic (flatter than normal)

LGD data have a bounded distribution, nearly symmetric and slightly platykurtic, so it is reasonable to be modeled via Beta distribution.

Beta distribution fitted	
Alpha	4.425694
Beta	5.4159389
p value of KS test	0.024755 < 0.05

By fitting Beta distribution, KS test rejects null hypothesis, suggesting that a single smooth fitted Beta distribution cannot explain all LGD behavior. Further diagnostics are based on the plot of actual and beta distribution, and related Q-Q plot, CDF comparison of actual and pure parametric LGD, plot of actual and pure parametric fitting, and related Q-Q plot (refer to Appendix).

Actual dataset vs Beta distribution, related Q-Q plot:

Plots show the Beta distribution captures the core shape well, with tiny deviations at extremes.

Actual dataset vs pure parametric, related Q-Q plot:

Plots show the fitted Beta distribution is well calibrated in the core, and bias is structural and confined to tails. There is no systematic curvature, and no slope distortion.

CDF comparison - Actual dataset vs pure parametric:

CDF plot shows very close alignment between empirical and Beta CDFs across most of the plot. The curves do not show global divergence, and deviations are localized.

Therefore, the hybrid approach of parametric core with light empirical adjustment is reasonable. Light empirical adjustment can fix minor deviations in tail regions, which is a localized tail correction, not a global distortion. Also, given that LGD tail outcomes are primarily driven by discrete legal and recovery regimes, empirical tail splicing is preferred for regime-dominated components. Tail cutoffs are set 7.5% and 92.5%, according to plots.

Based on the Inverse Transform Theorem or Probability Integral Transform (PIT), the semi-parametric LGD model is as follow:

$$F^{-1}(U) = \begin{cases} F_{emp}^{-1}(U), U \in (0, u_l] \\ F_{beta}^{-1}(U), U \in (u_l, u_u) \\ F_{emp}^{-1}(U), U \in [u_u, 1) \end{cases}$$

u_l : lower tail; u_u : upper tail

It reflects dynamics of LGD is regime-driven, e.g. good recovery regime, or loss-dominant regime.

Plot of actual dataset vs hybrid simulation, and Q-Q plot for tails (refer to Appendix) show hybrid tail-splicing implementation improves the tail overlap in the density. Another consideration is the correlations of LGD and EAD, and LGD and PD (predicted, corrected, by rating band).

Correlations of LGD and EAD		
Pearson	-0.008568	
Spearman	-0.004062	negligible correlation
Correlations of LGD and PD		
Spearman	-0.314286	model dependence

Abstract value of spearman with LGD and EAD is less than 0.1, so it is negligible correlation and may skip copula approach.

Abstract value of spearman with LGD and PD by rating band is more than 0.3, indicating model dependence. Since it is latent correlation, not systematic one, it does not need to jointly model with PD.

2) Lifetime LGD

To project multi-horizon LGD, it needs to consider mild maturity effects. LGD often shows slightly increase with horizon, and its effect is tiny, systematic, deterministic, not stochastic. Hence, this project models this maturity effect as a slow drift after generations of LGD baseline.

$$z = \log \frac{LGD}{1 - LGD}$$

$$z_t = z + \delta_t$$

$$\delta_t = \gamma \log(1 + t)$$

LGD: baseline from Monte Carlo simulation

t : horizon

For mild effect, $\gamma \in [0.03, 0.04]$

Given the bounded LGD, it will be no boundary violations with using logit space.

The simulation statistics of LGD for 1-5 horizons is as follow:

Horizon	Mean	Std	P05	P95
1	0.447177	0.152373	0.204484	0.693083
2	0.455804	0.152882	0.210810	0.701201
3	0.458067	0.153003	0.212490	0.703307
4	0.459823	0.153094	0.213800	0.704934
5	0.461259	0.153166	0.214875	0.706259

6.3.3 Exposure at Default (EAD) Projection

1) Baseline Simulations

Since the dataset has no information of loan type, recovery cash flow, collateral type and value, time in default, workout duration, utilization trends, repayment behavior, or scheduled principal payments, this project proposes conservative assumptions that all exposures are modeled as term loans with linear amortization.

This project starts to try lognormal fit, given skew is 4.946111, highly right-skewed.

$$\log(EAD) \sim N(\log(scale), shape^2)$$

the statistics of fitted lognormal is as follow:

Mean	1017651.40
Std.	1.133047
Shapiro-wilk test	
p value	0.293755 > 0.05 log-normal distribution likely
KS_test	
p value	0.783693 > 0.05, strongly fails to reject similarity

The results of test indicate that lognormal is a reasonable parametric choice. Also, plot of actual dataset and parametric, and Q-Q plot in log-space (refer to Appendix) suggest the lognormal core is well specified, with no curvature or systematic bias, and there is mild deviation in the extreme upper tail.

2) Multi-period EAD Paths

To be IFRS9- compliant and Basel IRB-consistent, EAD simulation should be aligned with PD migration. Concretely, EAD must be conditional on the same credit state

process that drives PD, whose migration is Markovian, generated by a rating transition matrix. Also, defaults stop the path evolution.

EAD may be decomposed to three components, based on cross-sectional structure: deterministic amortization backbone, rating-specific EAD shift, which is the systematic utilization shift due to credit states, and idiosyncratic exposure shock, namely bootstrapped residuals.

For the lifetime EAD, full simulation is three dimensions, obligors * paths * horizons. Given the computational workload of simulations, it is reasonable to collapse obligator dimension, and simulate representative exposure process by rating cohort, since EAD dynamics depends on rating transition, rather than obligator identity. Representative exposure paths are simulated per cohort, weighted by initial exposure. Also, rating migration and EAD dynamics remain fully consistent with the underlying transition matrix.

Such implementation transfers the representation of cross-sectional heterogeneity from obligor-specific paths to residual bootstrapping.

Another principle of rating-specific drift is that drifts are of monotonicity and consistence with deteriorating credit.

As for residuals, they are i.i.d., serially uncorrelated, with no auto-correlations.

The EAD projection is listed below:

Grade	Proportion
B	0.403834
BB	0.463355
CCC/C	0.132811

Initial rating = 'B'				
Horizon	Mean	P05	P50	P95
1	3,539,707	270,543	1,846,039	11,450,590
2	2,669,475	220,807	1,615,288	8,993,322
3	2,167,391	127,273	1,123,809	7,659,632
4	1,400,873	67,733	745,719	4,677,942
5	693,647	0	346,797	2,407,419

Initial rating = 'BB'				
Horizon	Mean	P05	P50	P95
1	3,509,639	317,074	1,981,933	12,405,340
2	2,790,983	130,783	1,526,238	9,083,520
3	1,788,357	0	988,503	6,392,462
4	1,205,995	0	617,137	4,833,713
5	554,210	0	281,662	2,005,757

Initial rating = 'CCC/C'				
Horizon	Mean	P05	P50	P95
1	3,523,346	289,228	1,960,448	12,361,410
2	1,763,507	0	667,316	7,256,910
3	984,339	0	0	4,380,213
4	467,879	0	0	2,419,989
5	249,225	0	0	1,157,442

6.3.4 Lifetime ECL Calculation

For lifetime ECL calculation, the summary of LGD, EAD, PD and their relations are showed below:

	Cross-section	5-year horizon	Shape
LGD	parametric core (fitted Beta distribution) + empirical CDF tail splicing	mild-maturity effect	simulation size: 10000 pre year, 10000*5
EAD	lognormal is fitted well. using rating cohorts as representations. Mean log_EAD as base + drift of rating + bootstrapped residuals	using S&P 1-year rating transition matrix as driver, aligning with PD migration, applying time-homogeneous Markov chain	simulation size: 1000 pre cohort, 3 cohorts, 3000*5
PD	predicted/corrected PD of 2024 based on training model, using survival power formula to calculate re-scalar factor, anchor S&P transition rating to predicted and corrected PD	S&P 1-year rating transition matrix as driver, matrix powering	simulation size: 3500 per year, 3500*5

PD~LGD	spearman_corr = -0.3143, by rating band, abstract value of spearman_corr > 0.3: model dependence	Dependence is explained by conditioning on systematic drivers, not by idiosyncratic joint shocks. PD and LGD share systematic risk. It is the latent correlation. Using Gaussian copula approach
PD~EAD	spearman_corr = 0.0286, by rating band, abstract value of spearman_corr < 0.1: negligible correlation	EAD and PD are aligned via rating migration. PD and EAD co-move. It is structural dependence, not residual dependence
LGD~EAD	spearman_corr = -0.0041, abstract value of spearman_corr < 0.1: negligible correlation	Exposure size is operational/contractual. Loss severity is recovery/collateral driven.

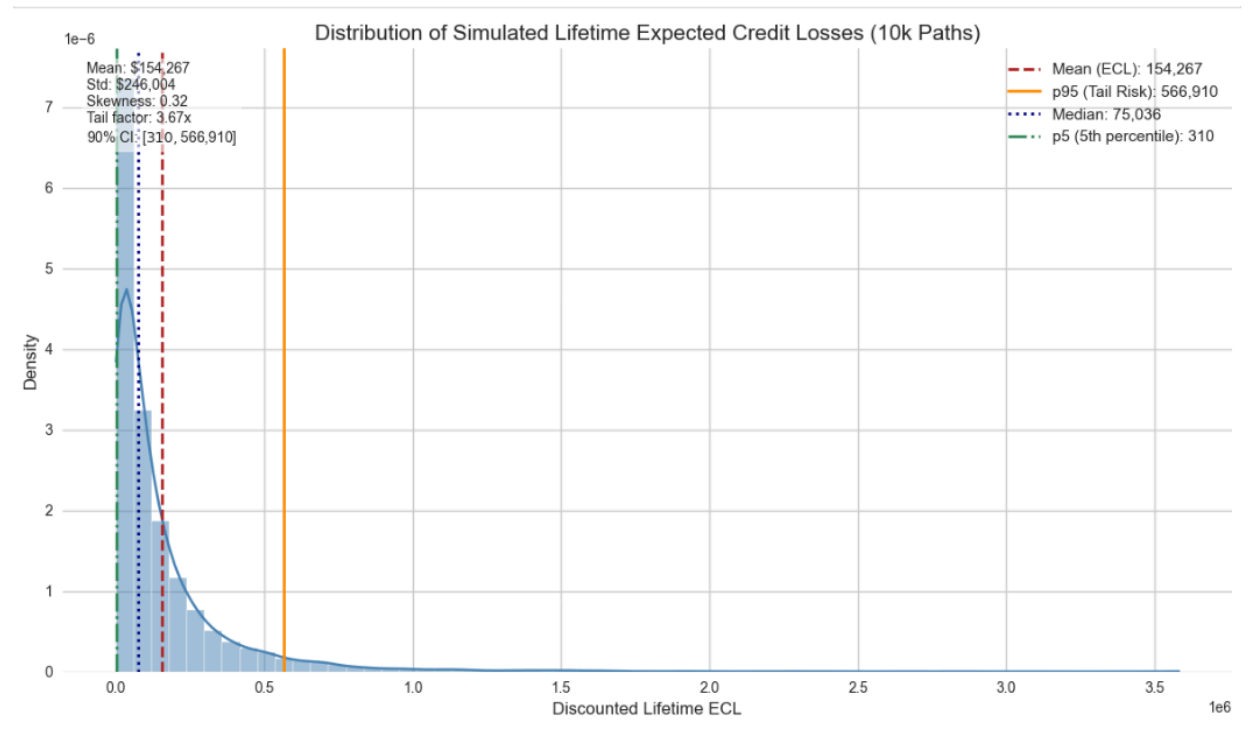
Based on the correlation of PD and LGD, Gaussian copula approach is applied to do resampling, and before it, Cholesky decomposition is performed to form correlated standard normal variable with a desired correlation matrix.

Due to limitation of Gaussian copula approach, its assumption of constant correlation and lack of tail dependence, it is unsuitable to tail risk and extreme risk analysis in market crashes. Student's t-copula or Clayton copula are more appropriate for economic capital and stress testing applications.

The statistics of lifetime ECL calculation is listed below:

Metric	Value	Interpretation
Expected Lifetime ECL	154,267	The "Red Line" on your plot; this is the accounting provision to be booked. The 50th percentile; 50% of your scenarios result in losses less than half the mean.
Median ECL	75,036	It reflects strong right skewness.
Standard Deviation	246,003	High volatility relative to the mean, common in credit portfolios with low default frequencies but high severity. The 95th percentile is 3.67 times larger than the mean, highlighting severe tail risk.
Tail Factor	3.67xmean	It is classic order of magnitude for credit portfolios.

Skewness	0.32	Moderate positive skew: quite mild, no explosive tail
90% Confidence Interval	[309.78, 566,909.79]	Extremely wide
95% Confidence Interval	[0.00, 780,644.22]	Extremely wide



The distribution of simulated lifetime ECL reflect the results are credible and internally consistent. They show the classic features of credit risk Monte Carlo output, i.e. right-skewed, mass near zero and occasional large-loss scenarios. 90% confidence interval aligns with p95 tail risk, which represent '1 in 20 years' event. Lower bound of 95% CI is close to 0, which indicates at least 2.5% of simulation paths have no defaults over the 5-year horizon, resulting in near-zero loss.

7. Future Work

7.1 XGBoost as a Diagnostic Tool

Deploy XGBoost with SHAP values as a non-parametric benchmark. If AUC of XGBoost is comparable to that of logistic model, it suggests that logistic model capture relevant structures.

XGBoost is appropriate to serve as a diagnostic tool to check whether the logistic model has missed crucial non-linearities or interaction effects, though it may suffer from interpretability, monotonicity enforcement and regulatory auditability.

7.2 Remediation of Band 6

Possible remedies may be Bayesian shrinkage, margin of conservatism (MoC), or platt scaling targeted only at the tail.

7.3 Scenario-adjusted PD

PD predicted and corrected from the model is Point-in-Time (PiT), and PD derived from matrix powering and from S&P rating matrices are through-the-Cycle (TTC).

Forward-looking information needs to be incorporated. The preferred approach is a Vasicek-style macro-overlay on the hazard rate. Macro adjustments should act on intensity, rather than marginal PD, to preserve the internal consistency of the multi-period term structure. An alternative approach is Asymptotic Single Risk Factor (ASRF) model.

7.4 Enhancement of EAD Modeling

At present, EAD is modeled via lognormal distribution. Possible enhancing tail approach is Generalized Pareto Distribution (GPD). In addition, drawdown scenarios may be considered as the overlay to current assumption of linear amortization.

7.5 Sensitivity Test of Lifetime ECL

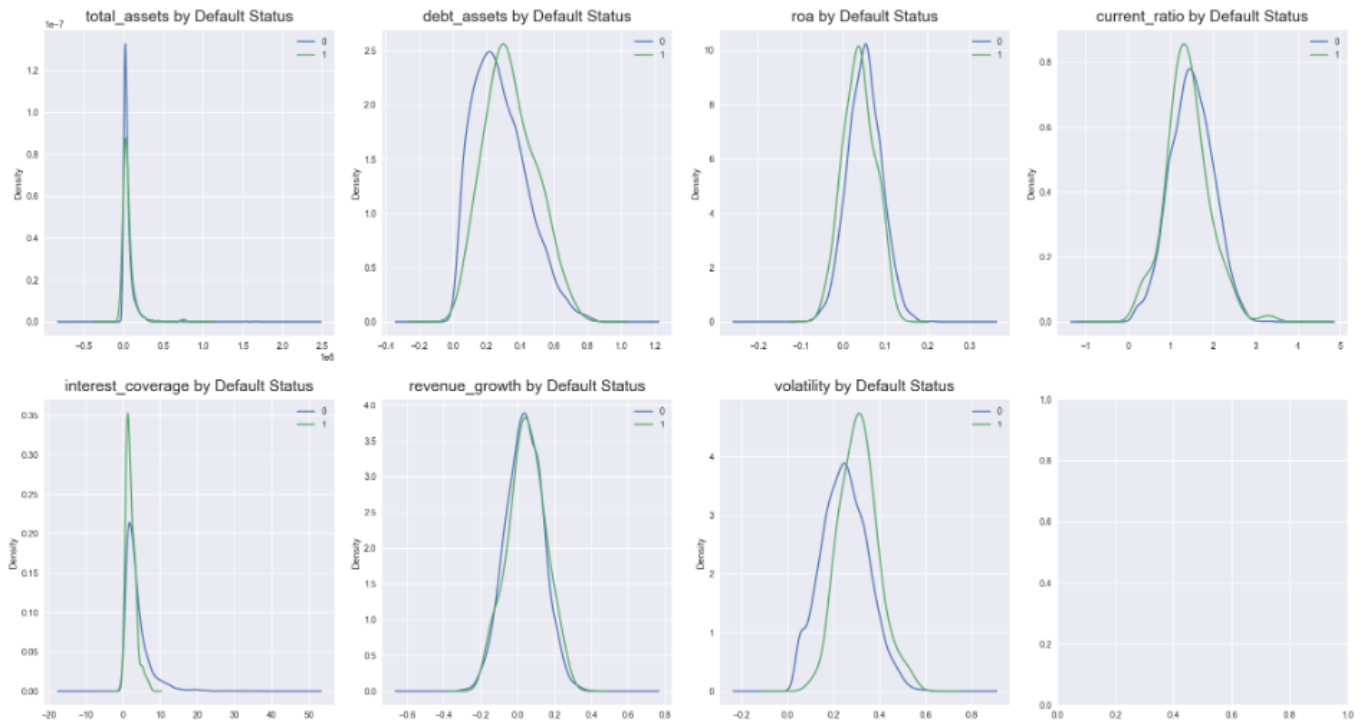
Stress testing and scenario analysis may be combined to lifetime ECL calculation, such as shock multipliers, macroeconomic mapping, and worst-case scenario.

7.6 Alternative Approach to Gaussian Copula

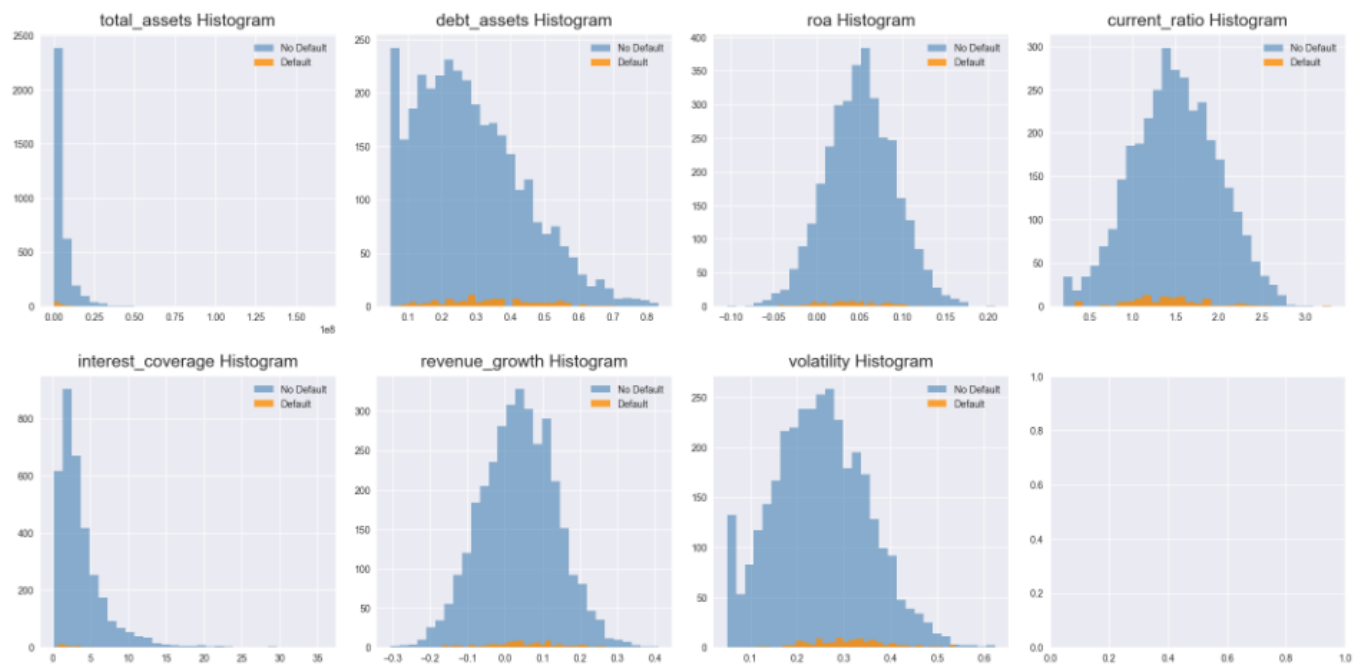
Constant-correlation structure may understate nonlinear tail dependence, especially in crisis regimes where default clustering, recovery deterioration, and exposure dynamics can become strongly state-dependent. A t-copula or Clayton copula would be preferable for tail-risk and downturn analysis.

Appendix

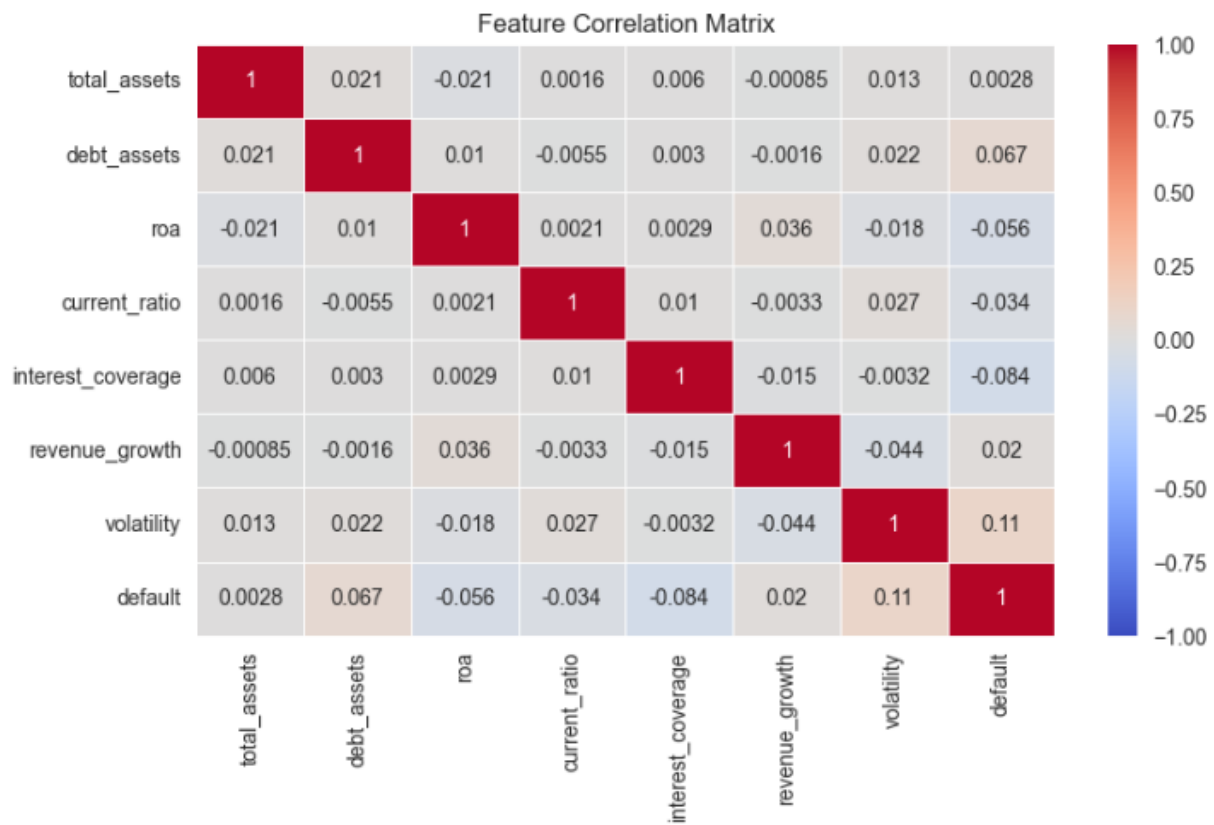
Probability density based on Kernel density of each feature grouped by defaults and non-defaults



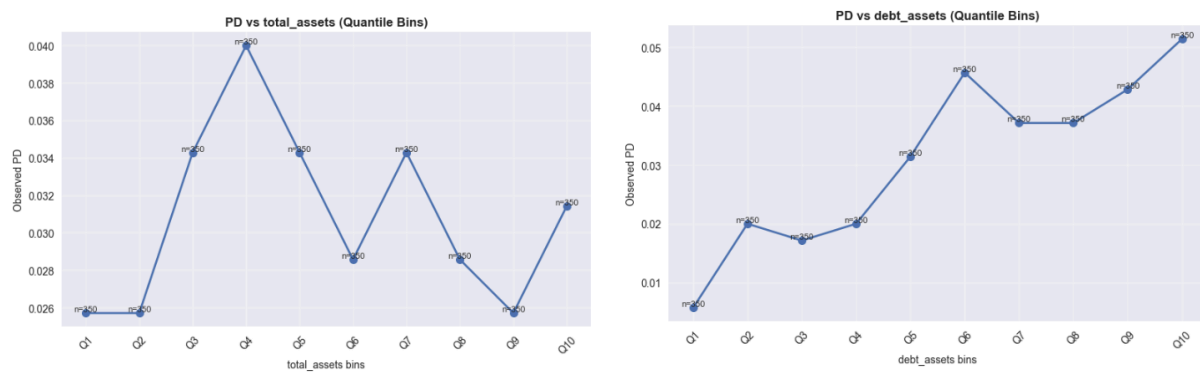
Histogram of each feature grouped by defaults and non-defaults

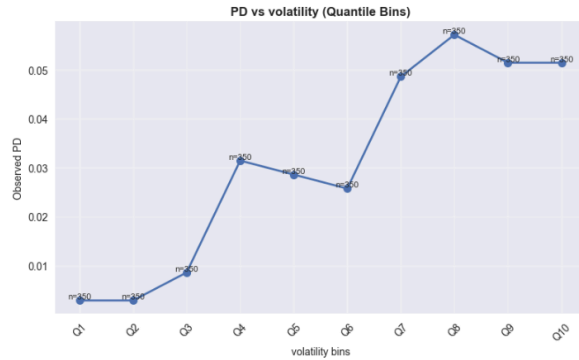
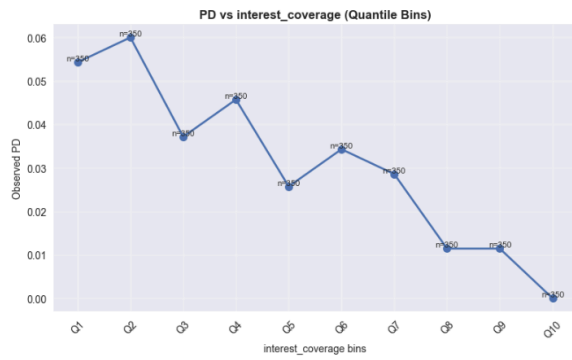
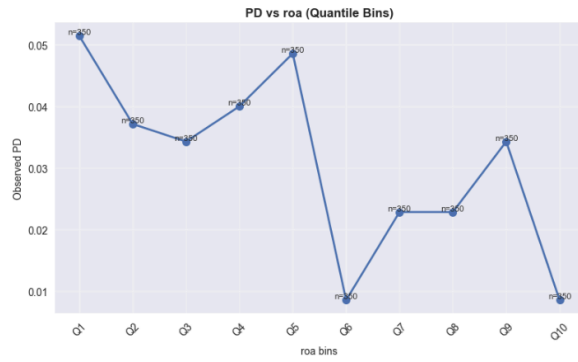


Feature correlation matrix

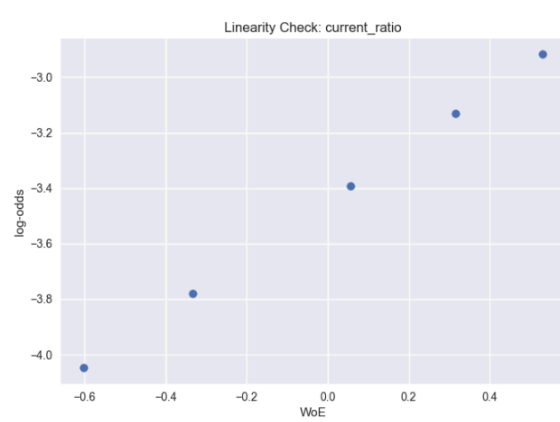
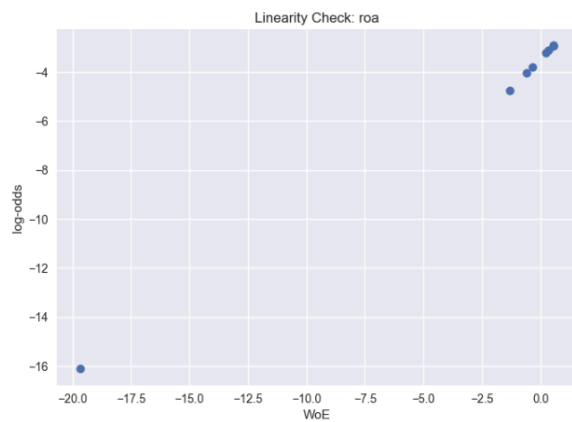


PD vs feature bins

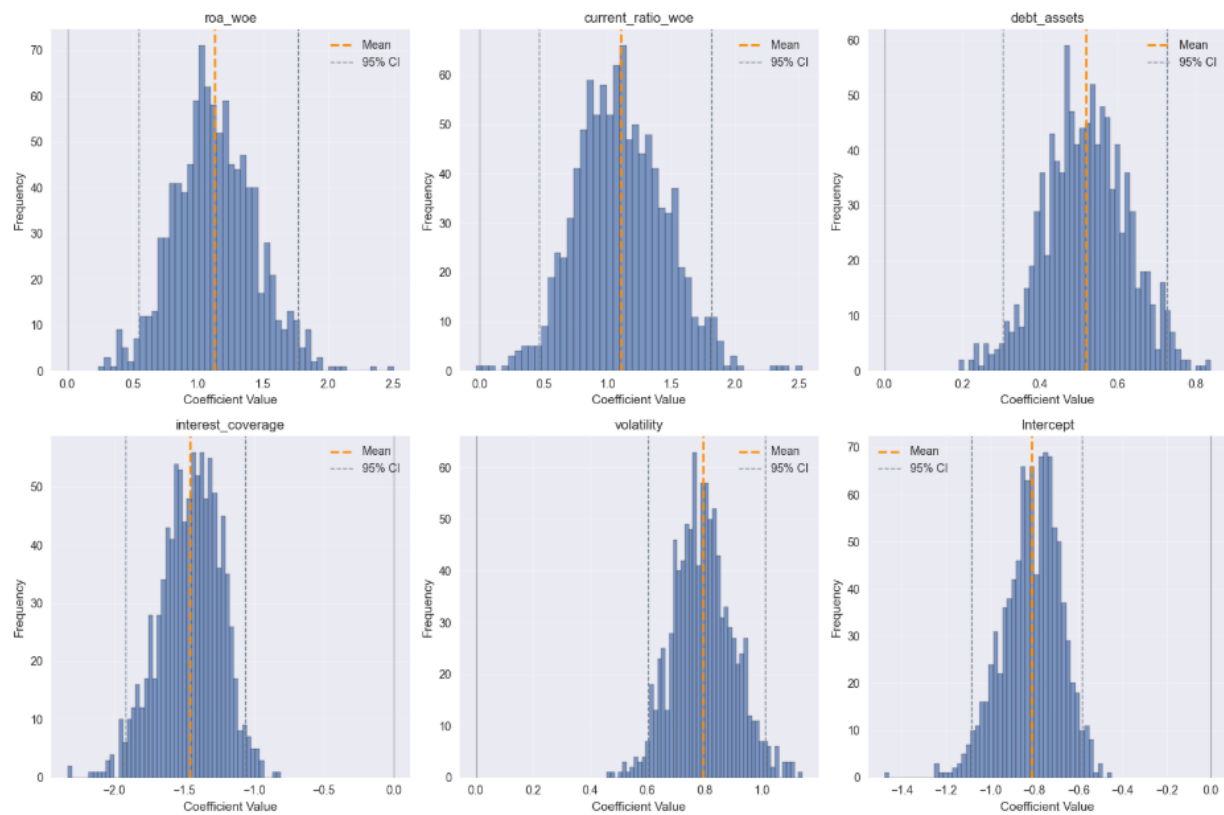




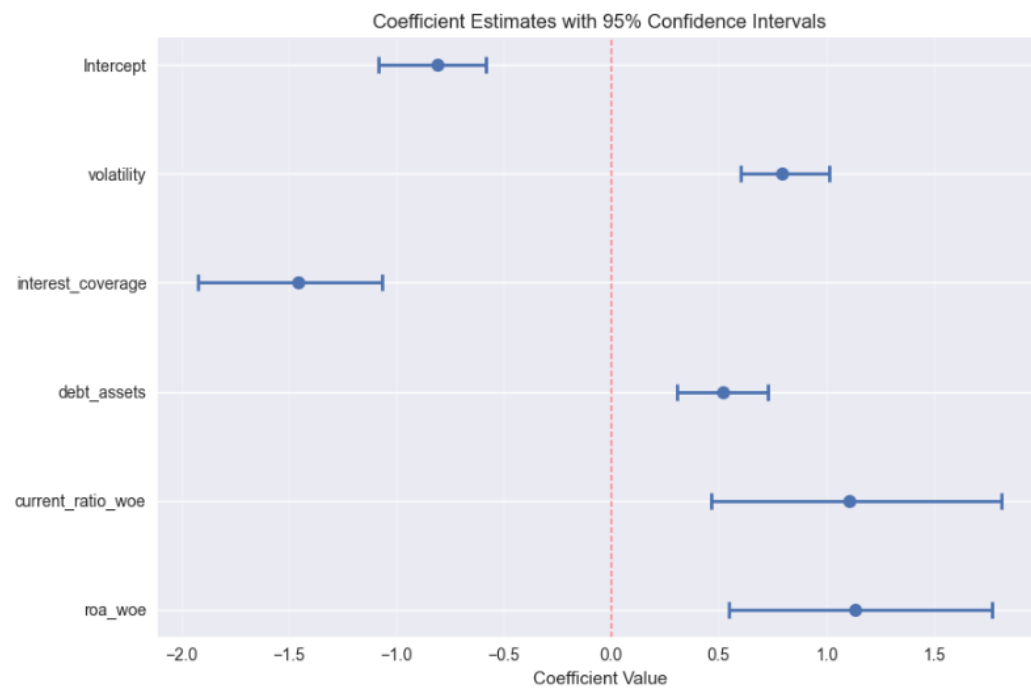
Linearity check



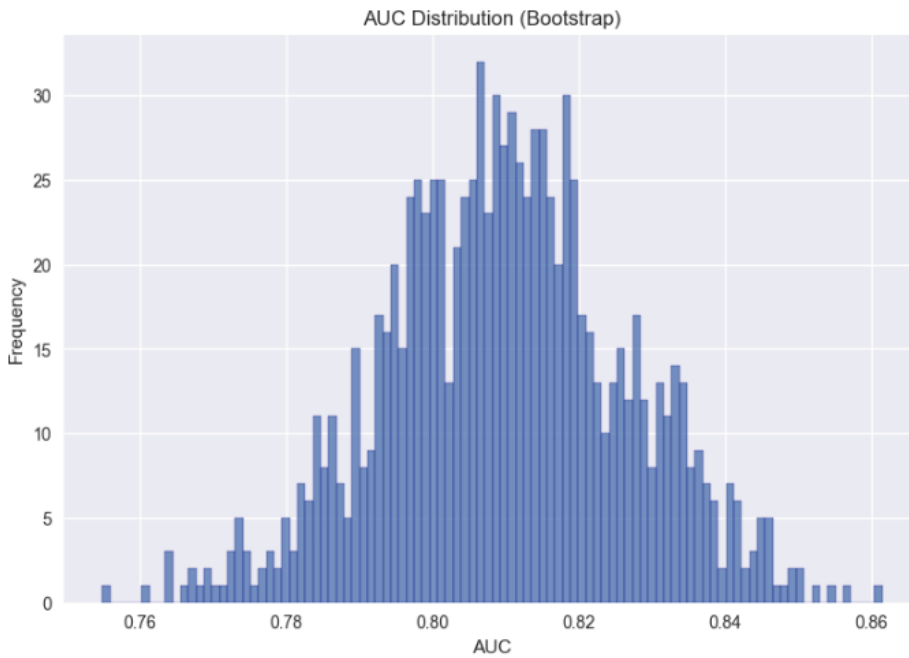
Bootstrap coefficient distribution plots



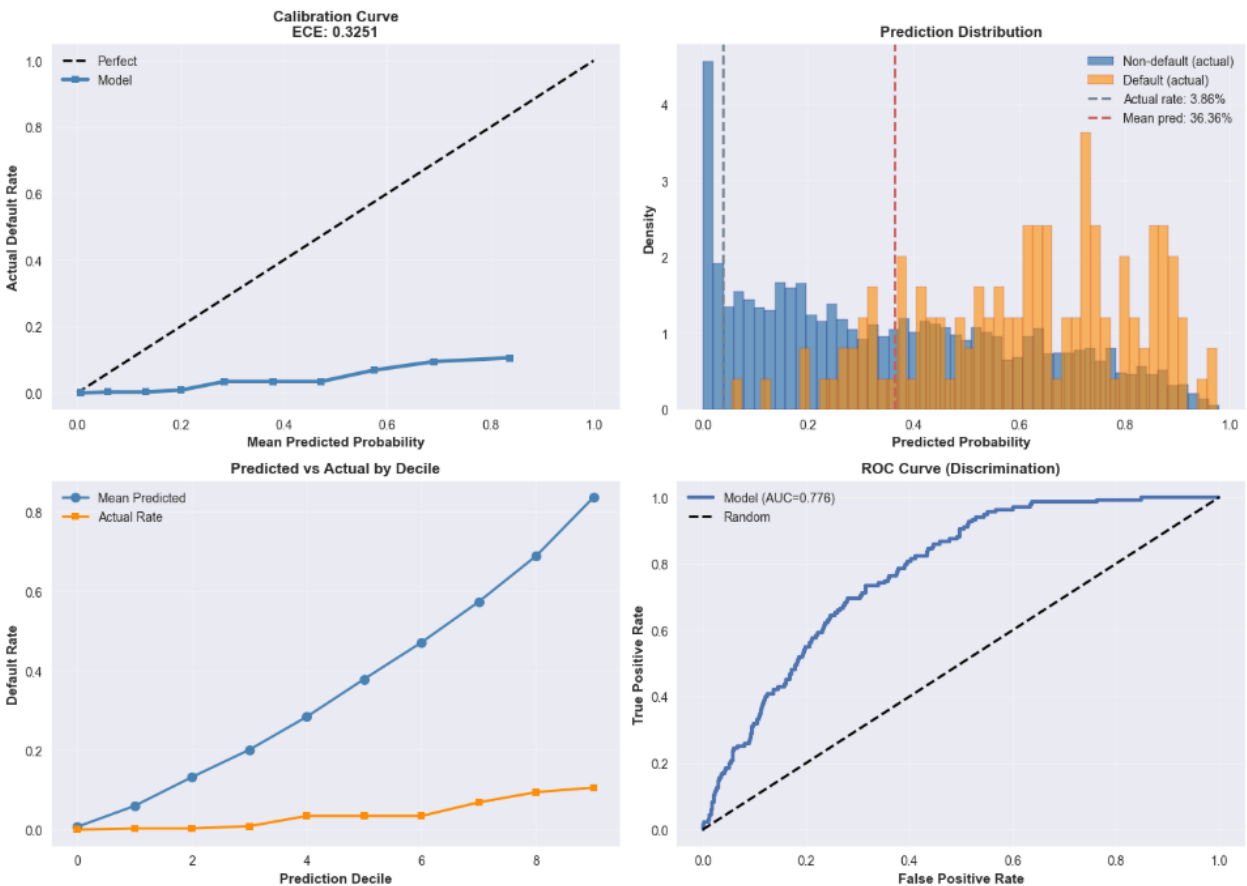
Confidence intervals



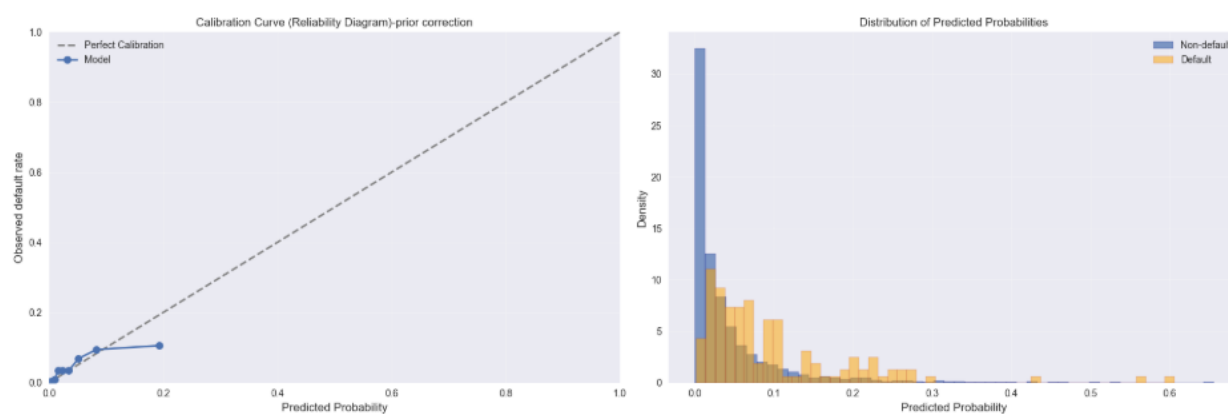
AUC Distribution



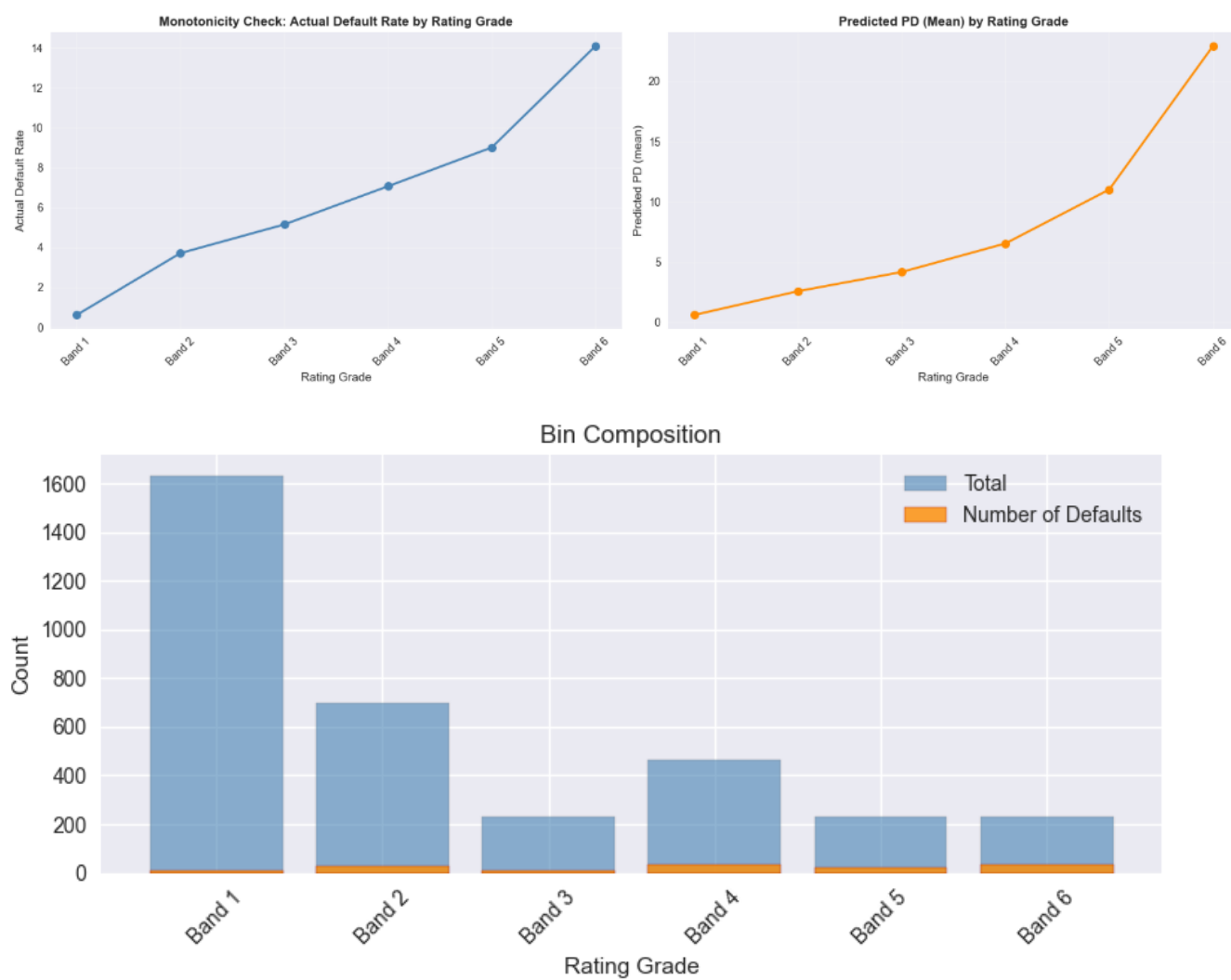
Calibration curves, prediction distribution, predicted vs actual rate, ROC curve

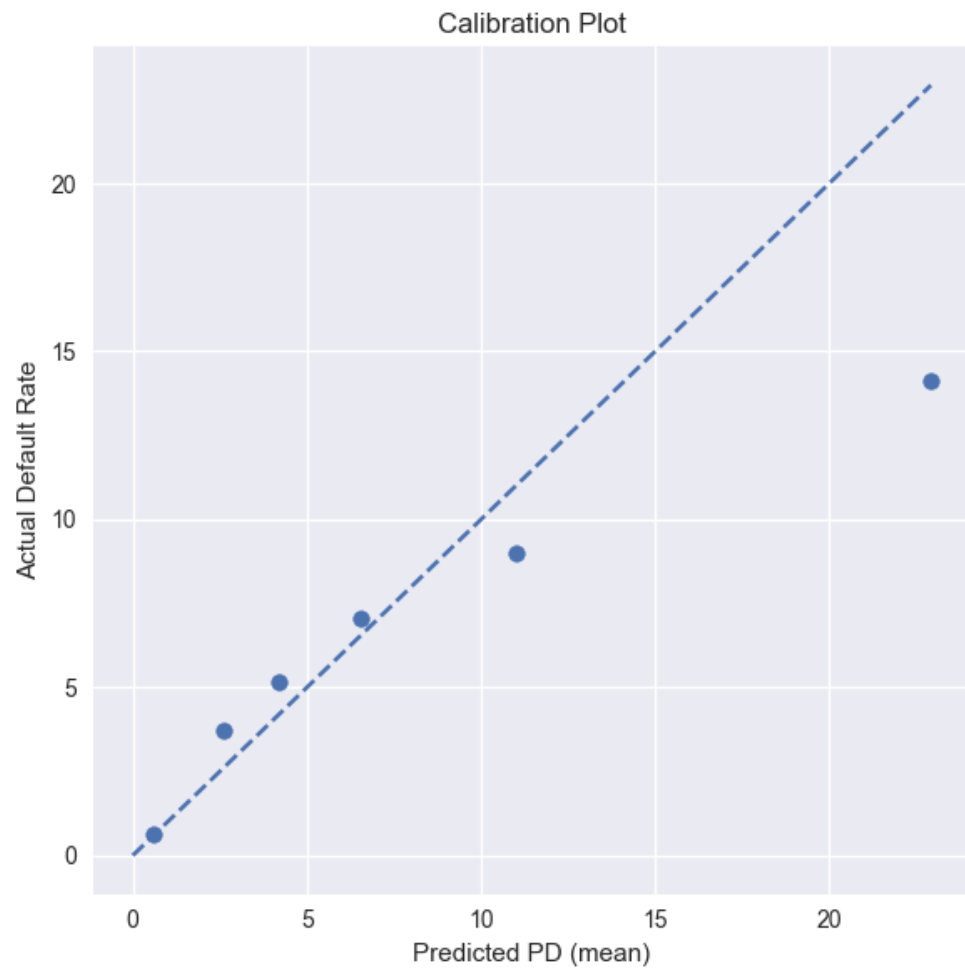
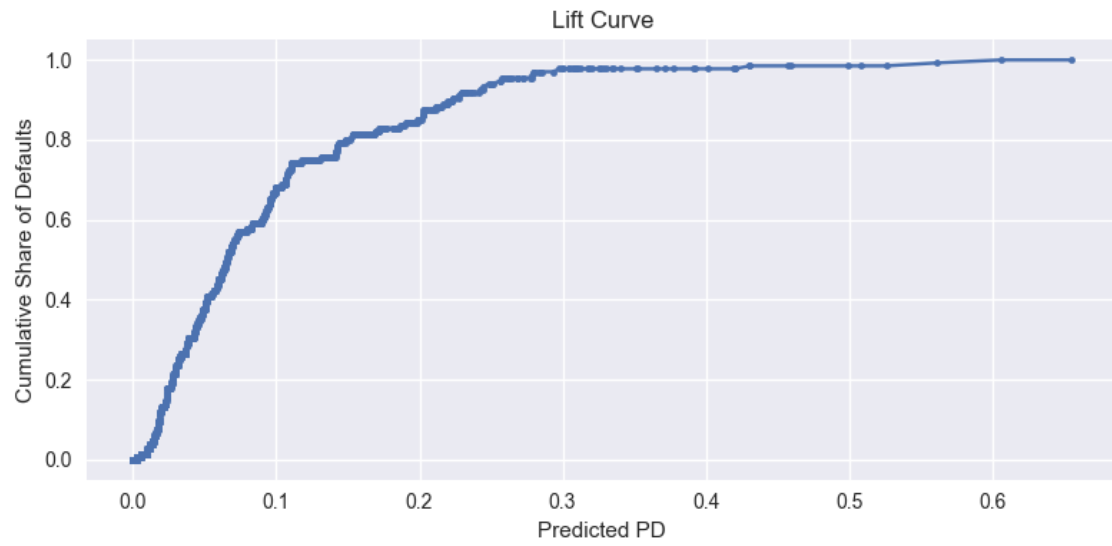


Calibration curve-prior correction and distribution of predicted probabilities



Rating grade





Average multiyear global corporate transition matrices

Table 33

Average multiyear global corporate transition matrices, 1981-2024 (%)

One-year transition rates

From/to	AAA	AA	A	BBB	BB	B	CCC/C	D	NR
AAA	87.28 (7.37)	8.92 (7.16)	0.51 (0.81)	0.03 (0.13)	0.10 (0.26)	0.03 (0.17)	0.05 (0.34)	0.00 (0.00)	3.08 (2.45)
AA	0.45 (0.52)	87.74 (5.18)	7.50 (4.19)	0.44 (0.65)	0.05 (0.18)	0.06 (0.19)	0.02 (0.06)	0.02 (0.07)	3.74 (1.69)
A	0.02 (0.08)	1.48 (1.04)	89.42 (4.06)	4.64 (2.21)	0.23 (0.36)	0.10 (0.22)	0.01 (0.06)	0.05 (0.10)	4.04 (1.71)
BBB	0.00 (0.03)	0.07 (0.14)	3.05 (1.58)	87.33 (4.09)	3.21 (1.67)	0.40 (0.62)	0.09 (0.19)	0.14 (0.23)	5.71 (1.52)
BB	0.01 (0.05)	0.02 (0.08)	0.10 (0.22)	4.44 (1.88)	78.89 (4.81)	6.25 (3.14)	0.50 (0.65)	0.56 (0.76)	9.23 (2.21)
B	0.00 (0.00)	0.02 (0.07)	0.06 (0.17)	0.14 (0.19)	4.47 (1.98)	75.18 (3.81)	4.79 (2.59)	2.93 (2.87)	12.41 (2.28)
CCC/C	0.00 (0.00)	0.00 (0.00)	0.07 (0.34)	0.13 (0.53)	0.40 (0.79)	13.18 (7.18)	45.07 (8.17)	26.12 (11.29)	15.03 (4.37)

Note: Numbers in parentheses are weighted standard deviations, weighted by the issuer base. NR--Not rated. Sources: S&P Global Market Intelligence's CreditPro and S&P Global Ratings Credit Research & Insights.

Table 34

Average multiyear global corporate transition matrices, 1981-2024 (%)

Two-year transition rates

From/to	AAA	AA	A	BBB	BB	B	CCC/C	D	NR
AAA	76.09 (10.45)	15.95 (10.50)	1.41 (1.44)	0.08 (0.24)	0.21 (0.41)	0.05 (0.23)	0.10 (0.40)	0.03 (0.17)	6.09 (4.21)
AA	0.78 (0.71)	77.18 (8.20)	13.30 (5.96)	1.14 (1.09)	0.16 (0.32)	0.12 (0.27)	0.02 (0.06)	0.05 (0.11)	7.24 (2.71)
A	0.04	2.64	80.17	8.14	0.59	0.21	0.03	0.12	8.06

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Table 34

Average multiyear global corporate transition matrices, 1981-2024 (%) (cont.)**Two-year transition rates**

From/to	AAA	AA	A	BBB	BB	B	CCC/C	D	NR
	(0.07)	(1.72)	(6.63)	(3.07)	(0.75)	(0.38)	(0.10)	(0.18)	(2.80)
BBB	0.01	0.14	5.61	76.75	5.17	0.88	0.17	0.38	10.88
	(0.07)	(0.22)	(2.49)	(6.59)	(2.14)	(1.00)	(0.27)	(0.58)	(2.45)
BB	0.01	0.03	0.24	7.82	62.60	9.44	0.93	1.77	17.16
	(0.05)	(0.09)	(0.44)	(2.94)	(6.92)	(2.95)	(0.82)	(1.99)	(2.99)
B	0.00	0.02	0.10	0.33	7.40	56.66	5.78	6.98	22.73
	(0.00)	(0.09)	(0.26)	(0.39)	(3.09)	(4.76)	(2.31)	(5.19)	(3.63)
CCC/C	0.00	0.00	0.10	0.33	0.80	16.95	22.18	35.79	23.84
	(0.00)	(0.00)	(0.39)	(0.98)	(1.02)	(7.01)	(7.53)	(12.54)	(5.99)

Note: Numbers in parentheses are weighted standard deviations, weighted by the issuer base. NR--Not rated. Sources: S&P Global Market Intelligence's CreditPro and S&P Global Ratings Credit Research & Insights.

Table 35

Average multiyear global corporate transition matrices, 1981-2024 (%)**Three-year transition rates**

From/to	AAA	AA	A	BBB	BB	B	CCC/C	D	NR
AAA	65.98	21.90	2.29	0.31	0.21	0.08	0.10	0.13	9.01
	(11.96)	(12.13)	(1.68)	(0.75)	(0.46)	(0.29)	(0.41)	(0.36)	(5.28)
AA	1.04	68.20	17.67	1.84	0.30	0.19	0.02	0.11	10.62
	(0.87)	(9.65)	(6.23)	(1.43)	(0.48)	(0.42)	(0.07)	(0.18)	(3.76)
A	0.05	3.46	72.29	10.62	0.98	0.34	0.07	0.20	11.98
	(0.09)	(2.16)	(8.08)	(3.15)	(1.00)	(0.56)	(0.13)	(0.26)	(3.61)
BBB	0.01	0.21	7.53	68.14	6.29	1.31	0.23	0.67	15.60
	(0.05)	(0.35)	(2.96)	(7.96)	(2.19)	(1.21)	(0.32)	(0.86)	(3.18)
BB	0.01	0.04	0.39	10.09	50.52	10.75	1.10	3.18	23.93
	(0.05)	(0.11)	(0.62)	(3.43)	(8.03)	(2.85)	(0.82)	(3.19)	(3.46)
B	0.00	0.02	0.14	0.59	8.79	43.11	5.39	10.60	31.37
	(0.04)	(0.09)	(0.38)	(0.69)	(3.46)	(5.02)	(1.96)	(6.57)	(4.50)
CCC/C	0.00	0.00	0.08	0.40	1.34	16.44	10.83	41.07	29.85
	(0.00)	(0.00)	(0.41)	(1.00)	(1.40)	(6.46)	(5.54)	(11.86)	(7.31)

Note: Numbers in parentheses are weighted standard deviations, weighted by the issuer base. NR--Not rated. Sources: S&P Global Market Intelligence's CreditPro and S&P Global Ratings Credit Research & Insights.

Average multiyear global corporate transition matrices, 1981-2024 (%)

Five-year transition rates

From/to	AAA	AA	A	BBB	BB	B	CCC/C	D	NR
AAA	50.08	28.63	4.73	0.85	0.23	0.16	0.08	0.34	14.91
	(12.30)	(13.29)	(2.73)	(1.50)	(0.45)	(0.40)	(0.27)	(0.57)	(6.79)
AA	1.30	53.71	23.71	3.24	0.49	0.33	0.04	0.28	16.92
	(1.02)	(9.76)	(5.12)	(1.91)	(0.62)	(0.56)	(0.09)	(0.37)	(4.52)
A	0.06	4.44	59.85	13.84	1.61	0.54	0.11	0.41	19.13
	(0.09)	(2.43)	(9.41)	(2.82)	(1.24)	(0.80)	(0.17)	(0.44)	(4.39)
BBB	0.02	0.33	9.72	55.52	7.11	1.79	0.30	1.42	23.79
	(0.06)	(0.49)	(3.00)	(8.72)	(1.73)	(1.38)	(0.35)	(1.37)	(4.14)
BB	0.01	0.06	0.77	11.78	35.27	10.74	1.10	6.00	34.29
	(0.05)	(0.15)	(0.89)	(3.08)	(8.26)	(2.02)	(0.84)	(4.71)	(3.64)
B	0.01	0.02	0.17	1.12	8.90	26.29	3.53	16.37	43.60
	(0.08)	(0.07)	(0.46)	(1.12)	(3.03)	(4.47)	(1.22)	(7.48)	(5.71)
CCC/C	0.00	0.00	0.07	0.57	2.34	11.59	2.54	48.18	34.70
	(0.00)	(0.00)	(0.40)	(1.47)	(1.93)	(4.73)	(3.10)	(11.38)	(7.89)

Note: Numbers in parentheses are weighted standard deviations, weighted by the issuer base. NR--Not rated. Sources: S&P Global Market Intelligence's CreditPro and S&P Global Ratings Credit Research & Insights.

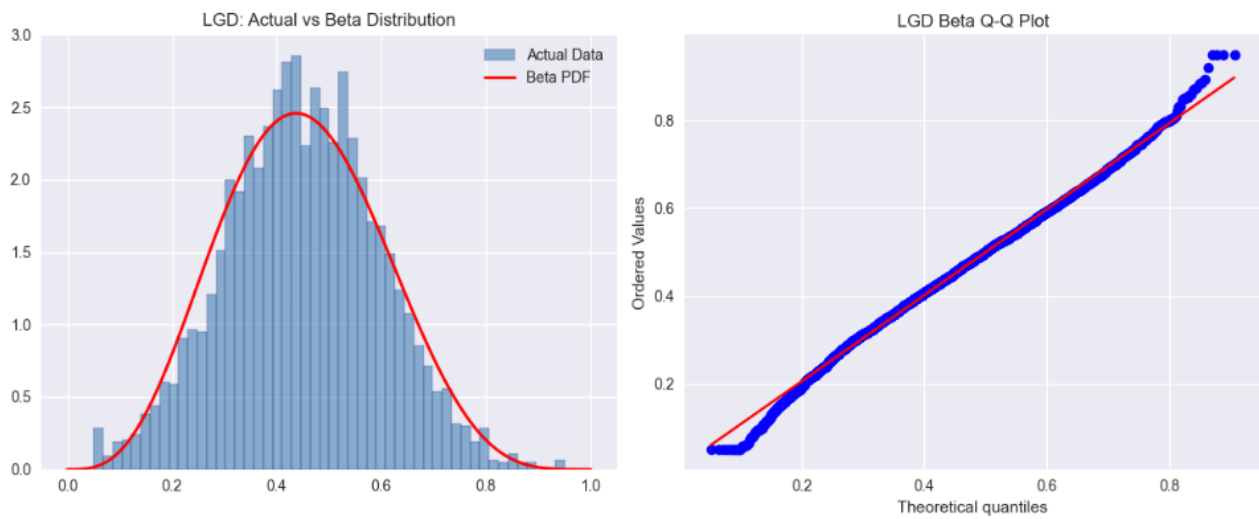
Table 37

Average multiyear global corporate transition matrices, 1981-2024 (%)

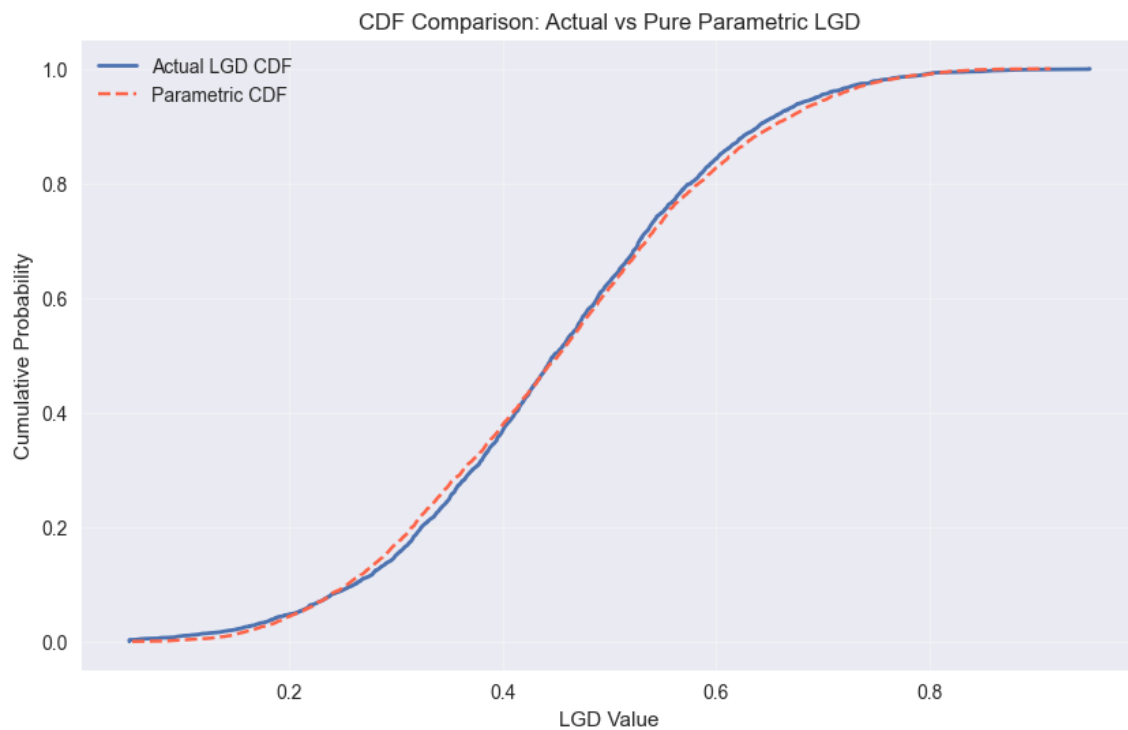
Seven-year transition rates

From/to	AAA	AA	A	BBB	BB	B	CCC/C	D	NR
AAA	38.53	32.69	6.77	1.50	0.23	0.18	0.10	0.49	19.49
	(11.32)	(13.65)	(3.05)	(1.85)	(0.46)	(0.42)	(0.30)	(0.70)	(7.27)
AA	1.37	42.89	27.08	4.30	0.65	0.33	0.03	0.47	22.90
	(1.03)	(8.68)	(4.11)	(2.01)	(0.62)	(0.49)	(0.08)	(0.53)	(4.66)
A	0.06	4.83	50.61	15.56	2.05	0.64	0.11	0.71	25.43
	(0.11)	(1.99)	(9.01)	(2.21)	(1.35)	(0.84)	(0.16)	(0.60)	(4.30)
BBB	0.03	0.43	10.83	46.69	7.03	1.96	0.30	2.15	30.58
	(0.09)	(0.54)	(2.50)	(8.19)	(1.02)	(1.17)	(0.31)	(1.74)	(4.42)
BB	0.00	0.06	1.07	12.25	26.17	9.59	0.90	8.50	41.46
	(0.00)	(0.15)	(1.00)	(2.81)	(7.30)	(1.99)	(0.60)	(5.63)	(3.30)
B	0.00	0.01	0.23	1.55	8.07	17.42	2.02	20.23	50.46
	(0.06)	(0.07)	(0.50)	(1.34)	(2.19)	(3.90)	(0.68)	(7.57)	(5.91)
CCC/C	0.00	0.00	0.20	0.81	3.13	6.93	2.10	48.83	38.01

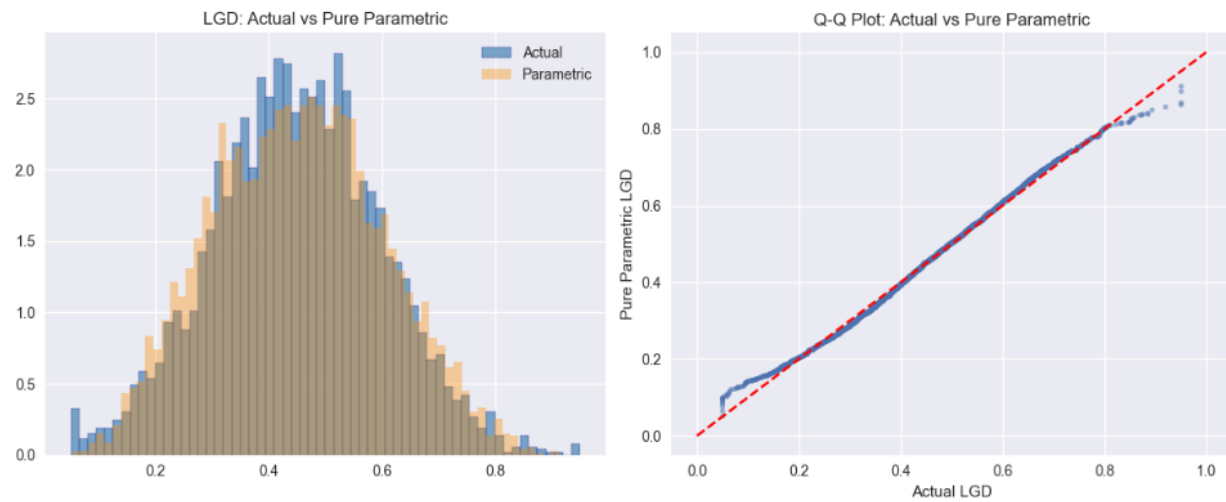
LGD: actual dataset vs Beta distribution



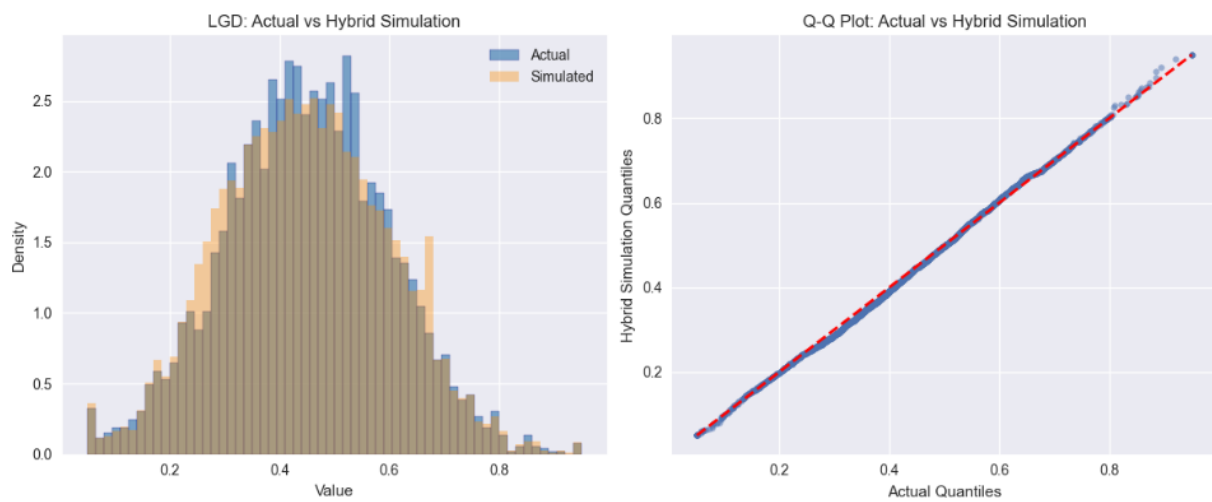
LGD: CDF comparison - Actual dataset vs pure parametric



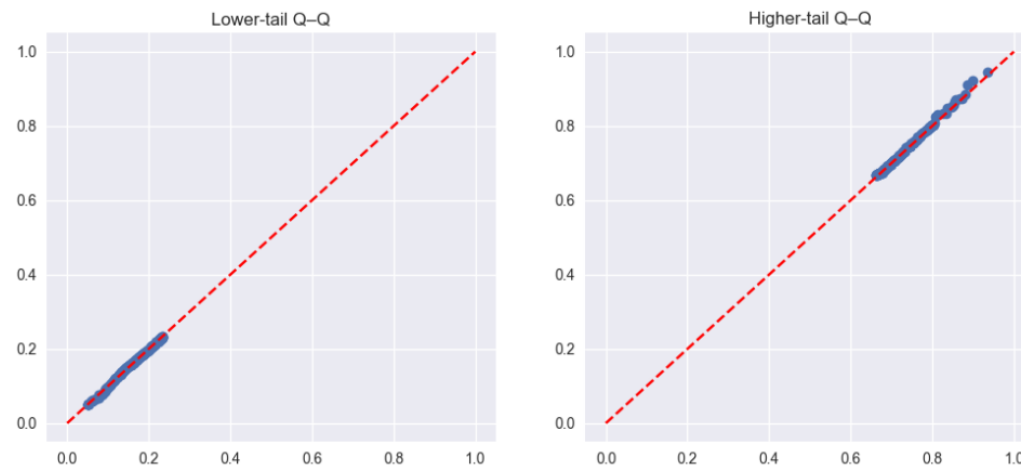
LGD: actual dataset vs pure parametric



LGD: actual dataset vs hybrid simulation



LGD: Q-Q plot for tails



Log (EAD): actual dataset vs parametric, Q-Q plot in log-space

